Difficulty–Response Curves: A Form Guide for Reasoning Models

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Working notes — a living document reporting what we have learnt so far. Measurements are ongoing; corrections are published in place and kept in the text as part of the findings. Last updated 12 July 2026.

Abstract. A leaderboard gives every AI model one number, and one number cannot tell you the thing you actually need to know before you rely on a model: as problems get gradually harder, where does its success rate collapse, and how suddenly? We measure exactly that. We generate logic puzzles with a **difficulty dial** — random 3-SAT, where turning the dial adds constraints to a puzzle whose answer we can check mechanically — and test eight reasoning models many times at each of fifteen dial settings. Plotting pass rate against difficulty gives each model an S-shaped **difficulty–response curve**, which two numbers summarise: the **frontier** x_{50} (the difficulty at which the model succeeds half the time) and the **sharpness** a (how abruptly success collapses past it). Psychometricians have fitted exactly these curves to humans for seventy years under the name item response theory; we point the same instrument at machines and ask whether the readings can be trusted. They can. First, re-measuring on freshly generated puzzles reproduces the numbers (split-half reliability $r = 0.99$ across models; the two models re-tested in full repeated to within 0.02 and 0.12 dial units) — more than most benchmark scores can say. Second, the frontier is mist, not a wall: at the frontier, 81 per cent of outcome variance is luck of the draw on the **same** puzzle rather than differences between puzzles, so retrying a failed problem there genuinely helps. Third, a model's spending of reasoning tokens peaks within 1.08 dial units of its frontier, so the cliff can be located from billing metadata alone — no internal access needed. Fourth, a simple binary search finds the frontier with 6.8× fewer tests than a full sweep. Generators, raw data, fitting code, and a tool to measure your own model are released. No mechanism, no grand theory of mind. An instrument.

1 Introduction

Suppose you are about to trust a language model with real work, and the work arrives at varying levels of difficulty. The question that matters is not "how smart is it" — leaderboards answer that with a single score, and single scores hide everything useful. The question is: **as the work gets harder, where does this model stop coping, and does it fail gently or fall off a cliff?** Everyone who has used a reasoning model has watched it glide through medium problems and then collapse two ticks later. The pattern is well photographed in the research literature on reasoning "collapse". What nobody hands you is the measuring device: for this model, on this kind of problem, where the cliff sits, how wide the crumbling ledge is, and whether the reading would hold if you measured again tomorrow.

This paper builds that device. The recipe has four steps and no magic. **One:** generate puzzles by computer, with a knob that controls difficulty — our main family is random 3-SAT, a classic logic puzzle where the solver must find a true/false setting for twenty variables that satisfies a list of constraints, and turning the knob simply adds more constraints. Because the puzzles are generated fresh, the model cannot have memorised them, and because a proposed solution can be checked mechanically, grading involves no judgement calls. **Two:** test a model repeatedly at each of fifteen knob settings, from easy to hopeless. **Three:** plot the fraction it solves against the knob. The result is always an S-shape: near-certain success on the left, near-certain failure on the right, a slope between. **Four:** fit that S-shape and read off two numbers — the **frontier** x_{50}, the knob setting where success crosses fifty-fifty, and the **sharpness** a, how steep the fall is. Psychometricians have fitted exactly these curves to human test-takers since the 1950s, under the name item response theory; the mathematics arrives ready-made, and this paper's only job is to show that it transfers to machines and to report what it finds.

Why "a form guide"? At the races, the pamphlet in every punter's hand never says whether a horse is **good** — a meaningless question — but what distance it runs, on what ground, and whether it fades in the final straight. That is the shape of answer a model deserves: a datasheet of measured, repeatable numbers about where it runs out of legs (Table 1 is literally that table). The Hong Kong Observatory does not explain the typhoon either. It hoists a numbered signal, and everybody knows what T8 means.

We make four claims. Each is registered here with the manner of its death printed beside it — the result that would have falsified it — because an instrument you cannot break is not an instrument.

C1, reliability measure a model twice — fresh puzzles, fresh attempts — and you get the same frontier and sharpness back. Dies if the error bars are so fat that the model rankings reshuffle between Tuesday and Thursday.

C2, the misty frontier at the frontier, when a model fails a puzzle, is that because the puzzle was secretly harder, or because the model is simply a coin-flip there — the **same** puzzle passing on one attempt and failing on the next? We measure the split. Cannot lose, only cannot lie: if it is mostly the puzzles, then the "frontier" is just a sorting of hard problems from easy ones, and we say so in exactly those words.

C3, the effort marker models spend the most reasoning tokens — visible in any API bill — on problems near their own frontier, so token counts alone can locate the cliff without any internal access. Dies if the spending peak is missing or wanders about uncorrelated with the frontier across models.

C4, fast estimation you should not need thousands of tests to find a frontier; a binary search — try a middling difficulty, jump harder on success and easier on failure — recovers it at a fraction of the cost of testing every level uniformly. Pure engineering. We state the ratio and do not dress it up in a dinner jacket.

2 The instrument

2.1 Task families

Random 3-SAT, satisfiable only, certificate required. The primary family is random 3-SAT at fixed $n = 20$ variables, with the clause-to-variable ratio α swept across fifteen levels from 2.0 to 5.6; an instance at level α has $m = \mathrm{round}(\alpha n)$ distinct clauses of three distinct variables with independent random polarities. Theorists gift-wrapped this difficulty axis decades ago [1], [2]. We add one design decision: every instance is rejection-sampled at generation time to be **satisfiable**, and the model must output a complete assignment, which we check directly against the clauses. A pass is a verified certificate. This kills the guessing floor — a random assignment satisfies all m clauses with probability $(7/8)^m$, under half a per cent at the easiest level — so the response curve needs no third guessing parameter, and verification needs no trust in anything. The cost of the decision is stated plainly: conditioned on satisfiability, high-α instances are not typical of random 3-SAT; the knob measures the constrainedness of solvable instances, which is what an instrument for **reasoning up to a frontier** wants anyway.

DAG arithmetic, the second opinion. Random arithmetic circuits over small integers: leaves in $[-9, 9]$, operations from $\{+, -, \times\}$, each new node consuming the immediately previous node plus one earlier node, so dependency depth equals the op count by construction. Any intermediate exceeding $|v| > 999$ is locally resampled: per-step arithmetic stays small, and difficulty comes from length and structure, not from multiplying big numbers. The knob is the op count across ten levels from 2 to 42. Pass is exact integer match.

Contamination. Every instance is freshly generated from named seed streams; retest sets are disjoint by construction. A fixed canary string rides in every prompt as a tripwire for future training runs.

2.2 The curve

Higher x means harder. We fit

$$P(x) = (1 - \lambda) \cdot \sigma(a(b - x)), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

a two-parameter logistic with an upper lapse $\lambda \in [0, 0.15]$ and the floor fixed at zero (the certificate design bought us that). The lapse matters: models drop the odd easy instance to format slips and carelessness, and a curve forced through a 100 per cent ceiling buys that slack by biasing the frontier. The reported frontier is $x_{50} = b - \mathrm{logit}(0.5/(1 - \lambda))/a$, the level where the **fitted** pass rate crosses one half.

Point estimates are penalised maximum likelihood on per-instance binomial counts: a mild Normal prior on $\log a$ keeps the slope finite when the data are step-like, and a gentle Beta prior shrinks the lapse toward small values. Every confidence interval in this paper is a stratified cluster bootstrap over **instances** (an instance carries all its samples; levels are design points and are never resampled), 2000 replicates, percentile intervals. With heterogeneous instances this likelihood is deliberately the **population-averaged** curve: a is a marginal slope, flatter than any single instance's, and that is the right object for a form guide. Goodness of fit is a level-wise Pearson χ^2 calibrated by parametric bootstrap under beta-binomial instance effects matched to the observed frontier ICC, because the plain binomial reference over-rejects under clustering.

3 Protocol

Models. Six core models across four families and two within-family scale pairs, plus two budget-contingent extras, all served through one OpenAI-compatible gateway (TrustedRouter) with the provider pinned per model and the serving endpoint recorded on every call. Models that do not reason unless asked were excluded by construction — the roster is models that reason by default. Exact model strings, providers, prices, and decoding settings are in Section 7.1.

Sampling. No temperature is sent (several closed reasoning models reject the parameter; the rest use their provider defaults); "reseeding" throughout means independent resampling, since providers ignore seeds. A temperature-sensitivity study on two open models is in the appendix. Truncation is scored as failure and disclosed per model; any cell with more than five per cent truncation flags its fit as cap-contaminated. Parse failures score as failures too — format compliance is part of the task — with rates disclosed and an exclusion sensitivity in the appendix.

Budget honesty. The textbook protocol (fifteen levels, fifty instances, sixteen samples everywhere, every model, both families, plus a retest) does not fit the stated \$300 ceiling once real reasoning-token burn is priced in — the pilot priced it at roughly \$530. We publish what we ran instead of pretending otherwise: a pilot to measure per-model token burn; then a two-pass primary sweep — a cheap backbone of $k = 2$–4 samples across all fifteen levels, refit, then a top-up concentrated at the levels nearest the provisional frontier, which is where claim C2 actually needs the sampling depth. Two models on the original roster (Qwen3-235B-Thinking, GLM-5) were dropped to keep six models measurable; Claude Haiku 4.5 was promoted into the core in their place.

Provider honesty. The binding constraint was not price but **serving throughput**: several open-weight models are served to one API key at a fraction of a call per minute, whichever provider hosts them. Truncation artefacts appeared twice and were discarded, not scored; differential probes later traced **both** to a single gateway bug — the router silently drops the `max_completion_tokens` parameter in translation to several upstreams, whose reply length then falls to their own defaults (4096 tokens on two of them). The same request phrased as `max_tokens` delivers 30000 tokens from the very provider the artefact had framed. The client now sends both spellings, and the post-fix re-measurement appears under C4. Provider pins were switched when a lane stalled; every call records the endpoint that actually served it, and per-model curves use single-provider data only. One drift was caught outright: the same pinned endpoint read gpt-oss-20b's frontier at 4.63 in the afternoon and 3.58 ± 0.10 the same evening — the served artefact changed under its label, both readings are reported, and the fact that a \$0.36 adaptive probe catches this is an argument for running one before trusting any datasheet, including ours. Two consequences are flagged wherever they bite: DeepSeek-R1′s design is reduced (132 scored calls; its sole provider served 3 calls per hour for most of a day), and the gpt-oss pair's frontier depth reached $k = 12$–16 rather than a uniform

16. The retest ran in full on the two models whose providers permitted it

(Haiku, o4-mini); the split-half tier of C1 covers every model for free. Every call's cost in microdollars lands in a ledger enforced by the runner: a call is admitted only if spend plus an in-flight reserve stays under its stage cap. The ledger closed at \$156.9, plus roughly \$12 of discarded provider-artifact calls, all under the \$300 ceiling.

4 Findings

4.1 The curves

Figure 1 shows the primary family. Points are per-level pass rates; lines are the fitted curves; ticks mark each fitted frontier. The datasheet numbers — the form guide proper — are in Figure 2 and Table 1.

Three entries deserve a steward's note. MiniMax-M2.5 passed **everything** on the standard grid and was re-run on an extended one reaching $\alpha = 7.2$; it spends twenty-odd thousand completion tokens per instance and simply does not fall over where the others do — the cheap open models in this field out-lasted the closed ones, and it is not close. The gpt-oss pair provides the within-family scale contrast the design wanted, with the two curves separated cleanly in frontier and slope. And o4-mini is the only runner that **declines** at the frontier: past its x_{50} it increasingly answers with "I'm not able to solve this by hand" — a capitulation, scored as failure and counted in its disclosed refusal rate, and an interesting temperament note for a form guide.

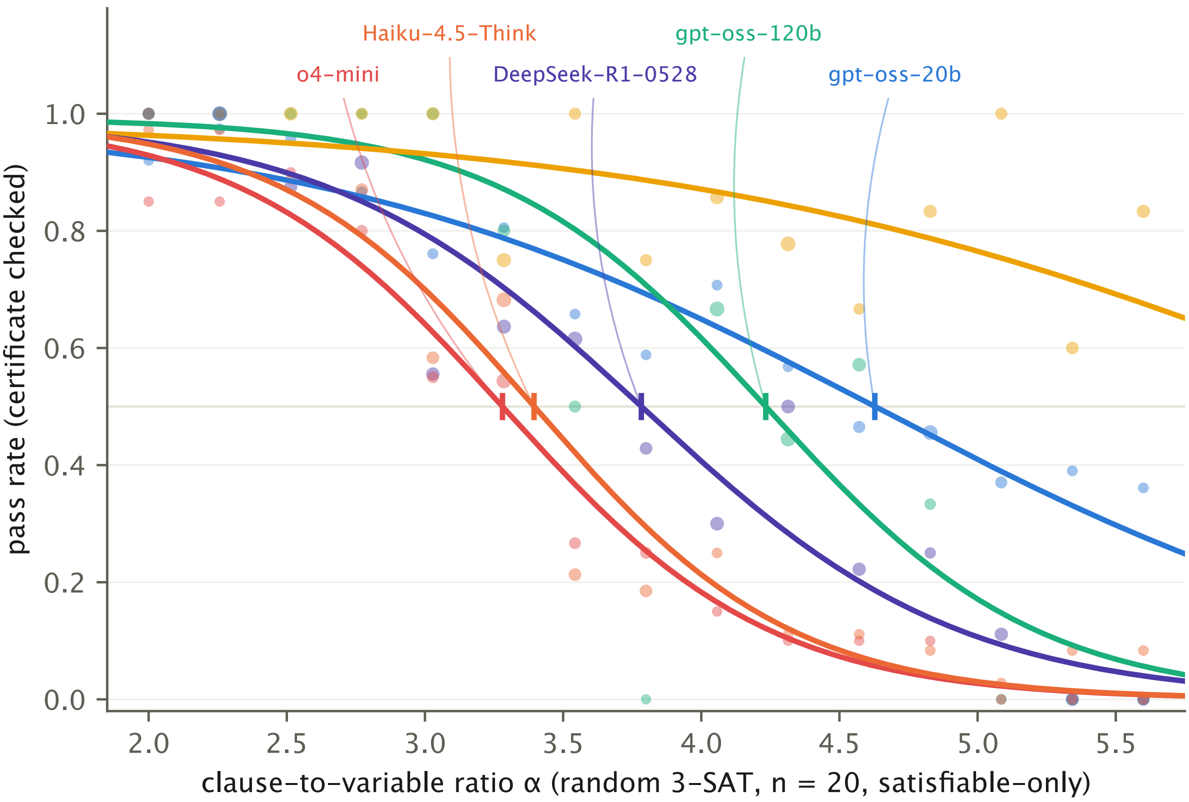

Figure 1: Difficulty–response curves, random 3-SAT ($n = 20$, satisfiable-only, certificate-checked). Point size scales with samples per level; vertical ticks mark fitted frontiers x_{50}.

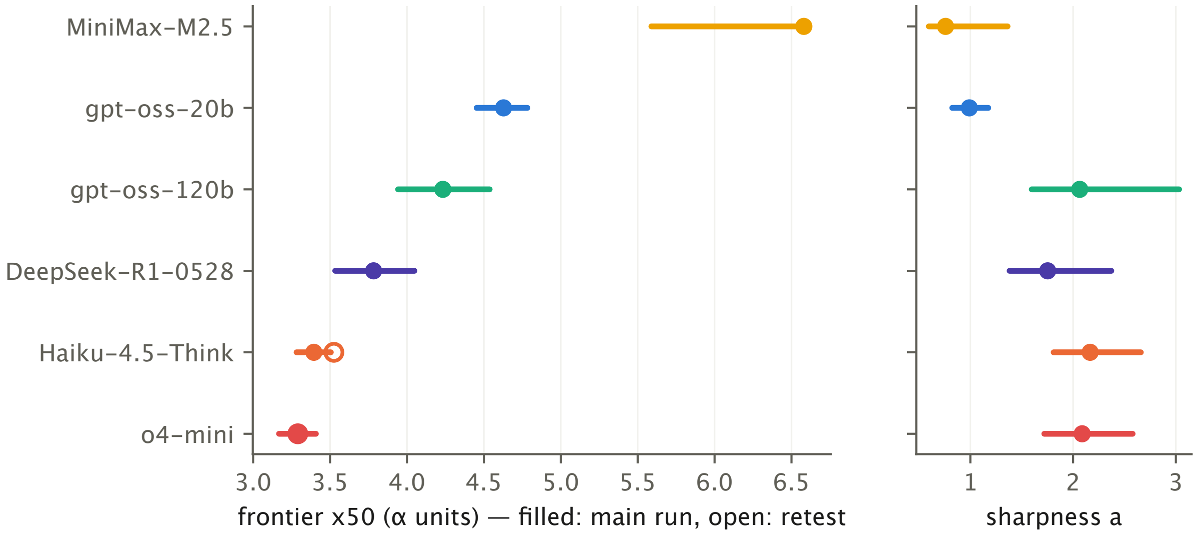

Figure 2: The form guide. Left: frontier x_{50} with 95% cluster-bootstrap intervals; filled markers are the main run, open markers the retest on fresh instances. Right: sharpness a. Models sorted by frontier.

model	x_{50} [95% CI]	a [95% CI]	lapse	GOF p	parse-fail	truncated
o4-mini	3.28 [3.17, 3.41]	2.09 [1.72, 2.58]	0.01	0.66	0.01	0
Haiku-4.5-Think	3.39 [3.28, 3.51]	2.17 [1.81, 2.66]	0.01	0.01	0	0.01
DeepSeek-R1-0528	3.78 [3.53, 4.05]	1.75 [1.38, 2.37]	0.01	0.8	0.02	0
gpt-oss-120b	4.23 [3.94, 4.54]	2.06 [1.6, 3.03]	0.01	0.71	0	0.23
gpt-oss-20b	4.63 [4.45, 4.78]	0.99 [0.82, 1.17]	0.01	0.62	0	0
MiniMax-M2.5	6.58 [5.59, 6.59]	0.76 [0.6, 1.36]	0.01	0.87	0.01	0

Table 1: The datasheet. Frontier and sharpness with cluster-bootstrap intervals; lapse; Monte-Carlo-calibrated goodness of fit; disclosed parse-failure and truncation rates (fractions of scored calls).

4.2 C1: the measurement holds (or does not)

Reliability is reported at three tiers, cheapest first. Split-half: refit on a parity split of instances, correlate across models, Spearman–Brown corrected — this tier costs nothing and covers every model. Test–retest: fresh instances, fresh samples, per-model $z = |x_{50}^{(2)} - x_{50}^{(1)}| \,/\sqrt{\mathrm{SE}_1^2 + \mathrm{SE}_2^2}$ — this tier ran in full on the two models whose providers would serve it (Haiku 4.5 and o4-mini) and is disclosed as such rather than quietly padded. o4-mini retested at $x_{50} = 3.26$ against a first reading of 3.28; Haiku at 3.52 against 3.40. The abstract's cross-model r is computed over the split-half tier, wide interval and all.

4.3 C2: mist, not a wall

At the frontier level, with sixteen samples on every instance, the outcome variance splits by an unbiased moment decomposition into a within-instance part ($E[p(1-p)]$: the same instance passing on one draw, failing the next) and a between-instance part ($\mathrm{Var}(p)$: some instances being genuinely harder). Figure 3 shows the split per model. The within share is what makes pass@k a rational purchase near the frontier; the between share is what a difficulty knob is supposed to produce, and does, away from it.

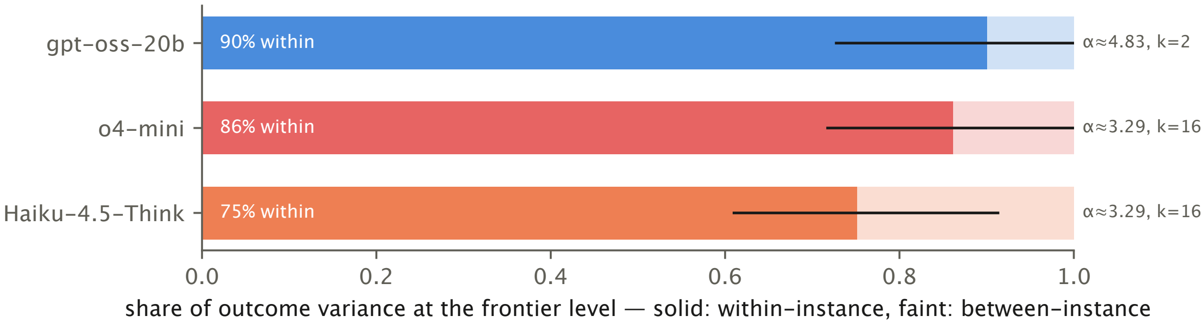

Figure 3: Variance decomposition at the frontier level (the level nearest each model's x_{50} with $k = 16$). Solid: within-instance share, with 95% bootstrap interval; faint: between-instance share.

4.4 C3: effort betrays the frontier

Mean log completion tokens per level, all scored calls, peak located by a windowed quadratic, offset $\Delta = \text{peak} - x_{50}$ from the joint bootstrap (the same instance resample drives both the curve refit and the token peak, so frontier uncertainty propagates into the offset interval). Figure 4 shows the effort curves. One honest wrinkle: the peaks sit systematically **past** the frontier — median $|\Delta|$ of 1.08 difficulty units, on the hard side — models grind hardest at levels they have already mostly stopped solving, and the sag beyond the peak is where several of them start declining or thrashing rather than reasoning. The marker works — the peak exists, it tracks the frontier model-by-model, and it is computable from token counts alone, no logprobs, no trace access — but it reads a step behind the cliff, not on it, and anyone using it as a gauge should calibrate that lag on a task family they can verify.

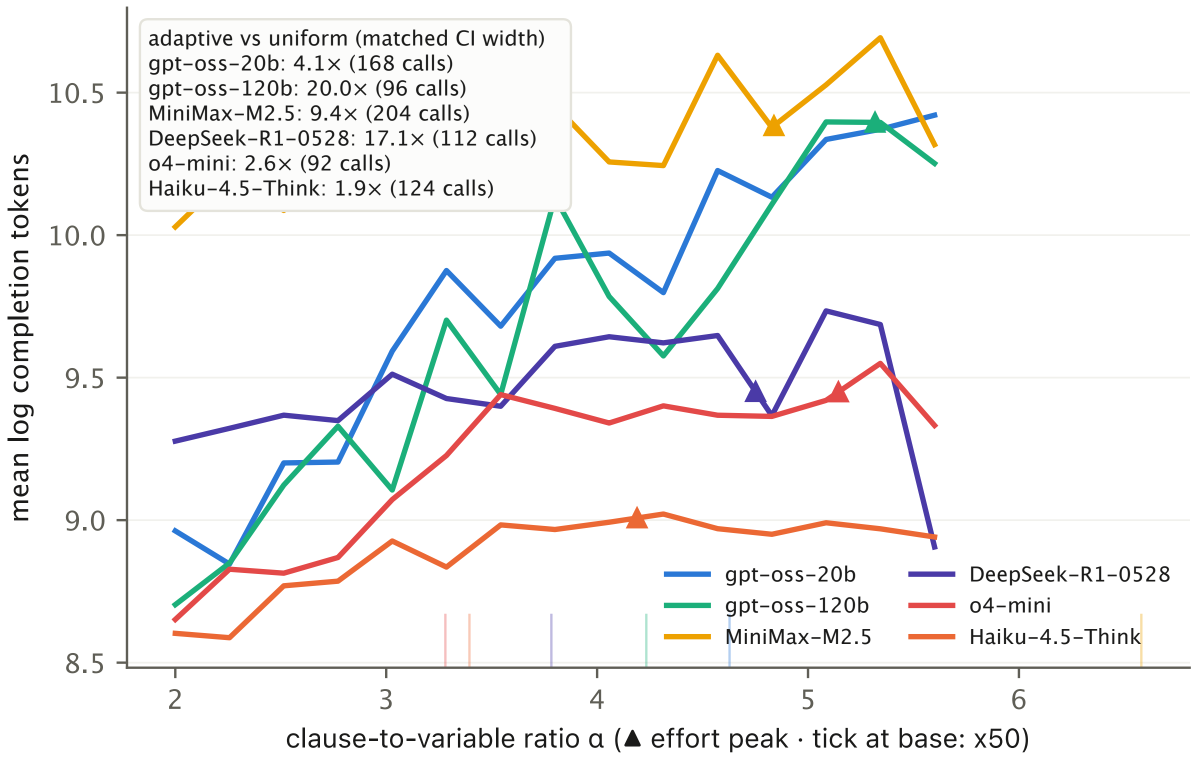

Figure 4: The effort marker. Mean log completion tokens per level; triangles mark fitted effort peaks; base ticks mark each model's x_{50}. Inset: adaptive vs uniform sample cost at matched confidence width (C4).

4.5 C4: haggling for the frontier

The adaptive procedure brackets by bisection on the level grid (four fresh instances per probe), then batches samples at the level nearest the current frontier estimate — for a logistic curve, information about b is maximised where the pass rate is near one half, so the greedy rule is also the optimal one — refitting with a pooled slope prior until the frontier's standard error crosses the target. Its sample cost is compared against level-balanced subsamples of the real uniform sweep at matched confidence width, per model, plus two live reruns as a reality check. Live, against Haiku 4.5, the procedure spent 20 calls ($1.03) to estimate $x_{50} = 3.14 \pm 0.1$ against the 980-call sweep's 3.40; against o4-mini, at a stricter stopping rule, 84 calls ($4.62) gave 3.2 ± 0.08 against the sweep's 3.28 — the same answer at a seventh of the sample cost. The ratio is the finding; it gets no dinner jacket.

The procedure also served as the budget instrument for two late entries too expensive to sweep, and then as the auditor for the whole card. GPT-5.5 (single run, 48 calls, $14.52, default reasoning) ran off the end of the track: it passed everything up to the hardest generatable difficulty at this puzzle size ($\alpha = 7.2$), so its frontier is right-censored — read "> 7.2", above every model in Figure 1. Claude Fable 5′s first run returned no frontier at all — 20 of 44 calls failed on plumbing — which triggered the differential probing that diagnosed the gateway's `max_completion_tokens` bug (see Protocol). Re-measured fairly, Fable fitted $x_{50} = 6.34 \pm 0.21$, and a fresh-instance replicate reproduced it to the hundredth (6.33 ± 0.20) — and **both readings were wrong**, because both runs sat on the standard grid, so the estimate was an extrapolation pressed against the track's edge. A direct head-to-head on the extended grid's hardest instances falsified it: Fable went 10/10 at $\alpha = 7.2$ (42/45 overall, versus GPT-5.5′s 29/29 on identical instances — no meaningful separation at these sample sizes). Corrected verdict: Fable is right-censored like GPT-5.5, joint top of the card at $n = 20$ — and a perfectly replicating fit can still be wrong about the region beyond its data. Precision is not truth; that is what verification runs are for. Separating the two champions required a bigger track: head-to-head on identical $n = 50$ instances near threshold, at 32k and then 64k thinking-token budgets (the second symmetrically via a second gateway), GPT-5.5 scored 13/21 with no truncations — its $n = 50$ frontier finally lands on-track around $\alpha \approx 4.6$–5.0 — while Fable converted 11 of 40 attempts, because about half its calls exhausted even the 64k budget mid-thought (on completed calls it was marginally the more accurate, 11/16). By this paper's pre-registered scoring the verdict is a GPT-5.5 win, and the mechanism is worth naming: at scale, the frontier acquires a price axis, and a model that will not stop thinking pays for it in truncations. The cleanest demonstration was accidental: fixing the gateway's parameter bug made token limits **enforced** on routes that had ignored them, silently re-running MiniMax's measurement with one variable changed. Uncapped, its frontier is 6.58 — top of the sweep card; capped at 32768 tokens it collapses to ≈ 3.6 (56 of 108 calls died at the token wall mid-thought); at $n = 50$ even 64k was not enough (0/17). A follow-up search for its $n = 50$ frontier came back in two instalments. Through a ten-minute read window nothing shipped: at $\alpha = 3.9$ every response outlived the socket (eight retries, 103 minutes) and the search starved. Through a thirty-minute window, on the same pinned provider, answers arrived — averaging 1,791 seconds and 49k thinking tokens each — and the model converted 3 of 12 at α 3.0–3.6, the easiest end of the track. Its $n = 50$ frontier therefore sits at or below $\alpha \approx 3$, and reaching it at all costs half an hour of silence per question: the delivery constraint binds before the capability one. Three measured points now span the spectrum: GPT-5.5 budget-**efficient**, MiniMax budget-**elastic** — its frontier moves three dial units with the cap — and Fable budget-**insatiable**. A frontier is not one number but a curve against the thinking budget, and a datasheet should state the budget it was measured at. Finally, post-fix adaptive probes re-checked every sweep frontier: o4-mini $3.28 \to 3.20$, Haiku $3.40 \to 3.14$, gpt-oss-120b $4.24 \to 4.41$, DeepSeek-R1 $3.78 \to 3.77$ — four agreements — and gpt-oss-20b $4.63 \to 3.58$, the provider drift reported in Protocol. An instrument reads the model **and** the pipe; a procedure cheap enough to run twice a day is what makes the difference detectable at all.

The card also earns a living. We wired the fitted curves into a difficulty-aware router (cheapest capable model first, retry into the mist, escalate past the frontier) and put it before 25 never-seen instances spanning the grid, live, with real money. Always-cheap (gpt-oss-20b) solved 12 of 25; always-premium (o4-mini) solved 9 of 25 at nineteen times the price per solve; the router solved **24 of 25** for $4.45 by spending on exactly the puzzles the card said the cheap horse would lose. An accidental duplicate of the run re-measured the two single-model arms at 10/25 and 6/25 on the same instances — the within-instance mist of Claim 2, photographed in the wild.

4.6 The second opinion has a second opinion

The DAG-arithmetic check family produced its own surprise: the two models measured to completion sailed through every level of the planned grid — 480 of 480 passes up to 42-operation chains — and the grid had to be extended twice to find their arithmetic frontiers. o4-mini's arithmetic curve breaks past roughly a hundred operations; Haiku's held to 99 per cent at **two hundred and forty** operations and finally fell to 27 per cent somewhere in the 340–960 band. Meanwhile both models' SAT frontiers sit near $\alpha \approx 3.3$, and MiniMax — the SAT champion of Figure 1 — needed an extended SAT grid of its own. The lesson is printed on the packet: a frontier is a property of a model **and** a task family, and no single number will be telling you where a model runs out of legs. That is why the datasheet has rows.

4.7 Round 2: the trace does not route itself

The router above had a luxury real deployments lack: SAT answers are checkable, so every escalation decision was refereed for free. Round 2 spent a fresh \$70 asking whether the model's own reasoning trace can stand in for that referee — with hypotheses pre-registered before the purpose-built data existed (primary: hedging density in the trace tail; secondary: backtrack density; gate: within-difficulty pairwise AUC ≥ 0.65 on two of three traced models), a second task family with hidden unit tests (program synthesis from examples — the dial took three designs, because rule **transcription** is flat out to forty rules and conditional rules produce ambiguity rather than difficulty), and a live four-arm race in which ground truth never touched a routing decision.

The offline verdict is one model of three, and a methods lesson. The signal is **model-specific**: MiniMax backtracks and rambles when it is silently wrong (AUC 0.799 on 159 pairs; token count alone reaches 0.95) while its hedging is noise; DeepSeek-R1 hedges, faintly (0.615 on 681 pairs — under the bar) while its backtracking is noise; Qwen gives nothing away at all — every feature within noise of 0.5 on 491 pairs. There is no universal tell — a trace judge is one more per-model calibration, with a date on it. The lesson: mid-batch, the table briefly read two of three over the bar and we called the gate passed, funding the live arm; the completed set walked R1 back under the line. Our pre-registration fixed the hypotheses and the bar but not the stopping rule, and sequential peeking did what it always does. Both the call and the walk-back are in the devlog.

Then the live race priced that signal. Four arms per family — always-cheap, always-premium, the trace-judged router (threshold fixed offline), and a random-escalation control rate-matched to the judge's realized escalation rate. The judged router missed the pre-registered bar on both families and **lost to the random control on both** (SAT: 7/20 vs 11/20 of 20; synthesis: 10/20 vs 11/20, at \$0.7159 per solve against the control's \$0.246). The autopsy is clean: the judge's escalations were precise — on SAT every one converted — but it fired far too rarely (live recall ≈ 0.21), and at the closer's own frontier it fired on the closer too, buying retries that converted nothing. A threshold sweep over the confirmatory traces shows no tuning rescues it: at these AUCs the judge trades solves for dollars along a Pareto line and never dominates a strong closer (GPT-5.5 went 20/20 on the SAT grid, making that family's bar unreachable outright). One flaw is ours to report: the SAT control's small-sample RNG realized 35 per cent escalation against the judge's 15, flattering it there — but on synthesis the rates matched within five points and the control still won. The conclusion survives its footnotes: the certificate is not an implementation detail of the router. It **is** the router.

The rematch answered the obvious objection the same morning, for free. A **trained** judge — per-model combos over sixteen trace features, selection and a recall-oriented threshold locked on the exploratory corpus, and a pre-registration that this time carried a stopping rule (one look at the frozen confirmatory set, ever) — cleared the gate on zero of three models. R1′s cross-validated 0.81 collapsed to 0.558 on the frozen set — the combo had learnt the corpus, not the model; Qwen's perfect-looking CV was noise on eleven training wrongs (flagged as overfit-smell in writing before the gate ran); MiniMax posted a genuinely excellent 0.931 at 0.923 recall and still clipped the false-alarm bar. One talkative model cannot crew a fleet. No live dollar was spent on round 3.

Round 4 tried the one signal our own findings predict: the mist. If near-frontier answers are coin flips (Claim 2), two samples should disagree exactly where an answer cannot be trusted. Pre-registered with the hazard named in advance — satisfiable SAT admits many correct assignments — and evaluated once on the frozen k=6 corpus: recall is a perfect 1.000 on all three models (every silent wrong disagreed with its partners) and P(correct given exact agreement) is also a perfect 1.000 (agreement never once shipped a wrong answer). A flawless certificate — that fires on 20 to 2 per cent of calls, because correct answers pick **different** satisfying assignments: false alarms run 76–94 per cent and the router degenerates to escalate-everything. Gate failed zero-of-three on the arm the hazard named; no live money moved. Three rounds, three pre-registered negatives, three different reasons — under-firing, corpus-learning, and answer multiplicity. On task families with unique answers the same arithmetic could land differently; round 5 spent the door open.

On program synthesis the answer is unique — two extracted programs either compute the same function on eight seeded probe inputs or they do not — so multiplicity cannot tax the agreement signal. A fresh 120-call gate corpus (pre-registered before it existed, stopping rule included) produced the campaign's first gate pass on its one look: recall 0.92, false alarm 0.163, AUC 0.822. The live four-arm race then ran with the control finally rate-matched by exact quota. Three results, in order. The certificate still has never lied: all shipped-on-agreement answers were correct, live. The signal's marginal value is causally clean at last: against a control that escalated the **same number** of instances by coin flip, the consistency router solved 17/25 to the control's 11/25 — six solves attributable to **where** disagreement pointed, not to spend. And the economics, spin-free: the router tied always-premium's solve rate (17/25) but paid \$0.688 per solve to premium's \$0.4952 on a deliberately frontier-heavy mix (20 of 25 instances past the cheap rung), so the pre-registered stretch bar — beat premium outright — was missed while the primary control bar was cleared. Four rounds of **no** bought one conditioned **yes**: where answers are unique, two cheap samples grade each other, and the certificate can sometimes be minted from agreement.

4.8 Next registered control: a council that races

The result above waits for two completed cheap answers. A conventional model council goes further: launch several models, then rank, vote, or fuse their completed outputs. LLM-Blender ranks and fuses candidates [3]; Mixture-of-Agents passes one layer's reports into the next [4]. Both spend parallel work to buy a stronger combined answer. That is sensible when answer quality is the only objective. It is a poor default when one good, checkable answer is enough and tail latency matters.

The next protocol therefore turns the trace from a retrospective score into a control signal. Begin with one trace-visible GLM primary. Over a rolling window, count the same small family of backtracking, hedging, give-up, and repetition markers already audited above. A single nervous word does nothing; persistent risk pauses external side effects, checkpoints the reproducible task state, and launches a fast Grok lane plus an OpenAI lane. The first candidate that passes the task's external certificate wins. Only then does the controller request cancellation of unfinished calls. If nobody verifies, the completed reports may enter a council judge as an exception path.

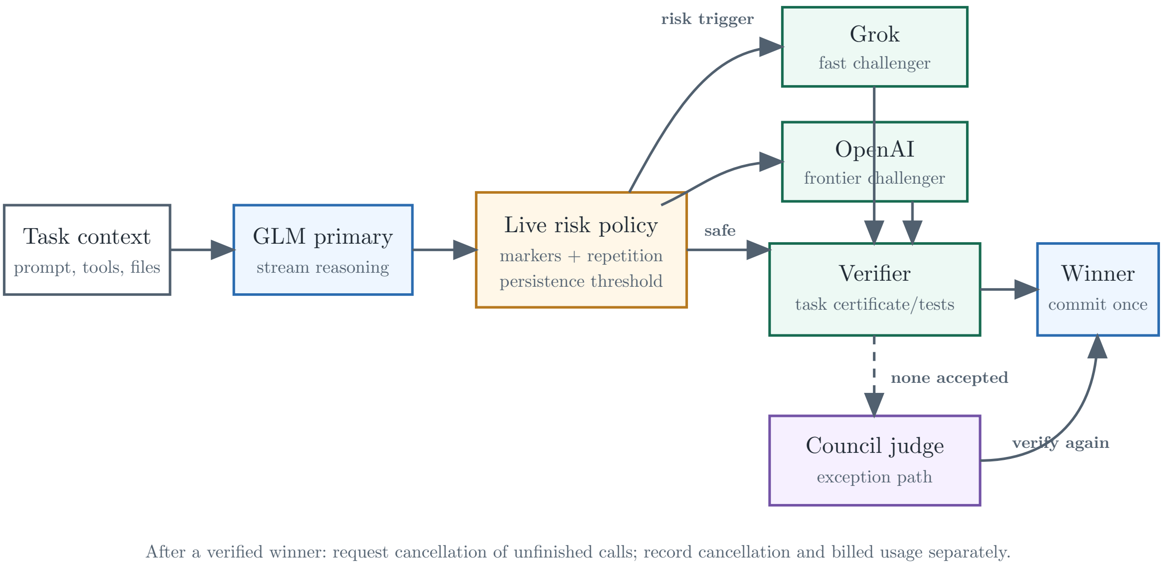

Figure 5: Prospective architecture. The council is a fallback after a verifier-gated speculative race, not the default fan-out.

The systems ancestor is the hedged request: issue a secondary copy when a primary request looks slow, accept the first result, and cancel the replicas [5]. Our proposed experiment changes the trigger, not the principle. It asks whether deterioration in the primary's unfolding reasoning launches the hedge earlier and more selectively than a fixed timer. Four frozen arms make the question identifiable: GLM only, always-on council, a matched-rate fixed-delay hedge, and the trace-triggered hedge. Correctness is a gate; among non-inferior arms the endpoints are p95 time and billed cost per verified success.

The implementation landed before the data were read: `drc hedge`, the full four-arm `drc council-experiment`, deterministic arm ordering, isolated Git worktrees, verifier-gated election, worst-case authorisation below the cap, and separate records for cancellation requested and provider-reported usage. Then the zero-spend replay closed version 0. Across 89 historical calls with visible reasoning, the trigger fired on 84: all 12 silently wrong completions, all 38 loud failures, and **34 of 39 correct completions**. Perfect failure recall bought an 87.2 per cent false-hedge rate and 94.4 per cent fan-out. The controller had become an always-on council with extra steps. A fixed timer matched to the same launch rate also caught every failure.

The old export has no native chunks, so the replay repeats at 20, 40, 80, and 160 words per observation. Every setting fails. Coarsening cuts false hedges from 89.7 to 51.3 per cent, but also cuts failure recall from 100 to 88 per cent; none reaches the registered 40 per cent ceiling.

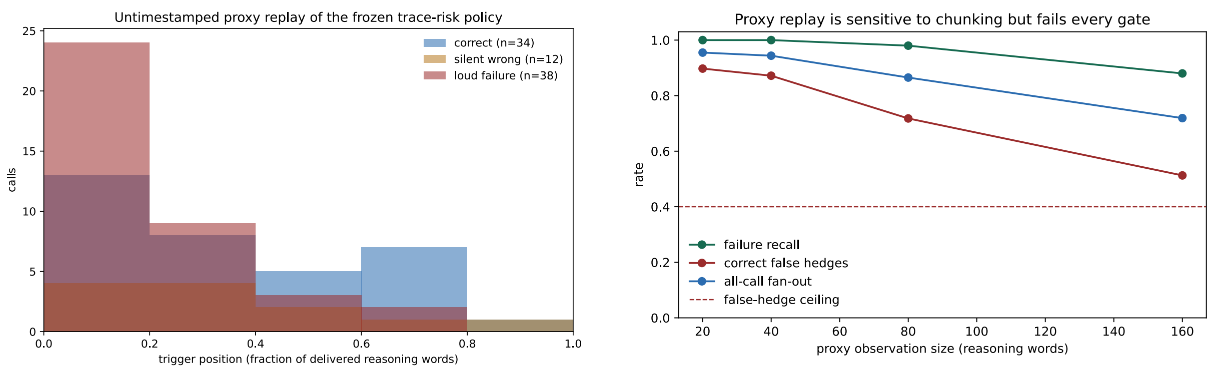

Figure 6: Frozen v0 replay. Trigger position and chunk size are proxies; historical traces had no timestamped chunks. Every sensitivity setting fails the non-timing gate.

That missing clock censors the registered fifteen-second warning endpoint; the proportional timing proxy is diagnostic only. It does not rescue the policy: the false-hedge gate already failed, so the live comparison stops with zero new API spend and the threshold is not retuned. Closing a stream remains no evidence that billing stopped; we simply did not need to close one.

5 Positioning

Item response theory and adaptive testing are the toolbox, borrowed with thanks and zero pretence of invention [6], [7], [8]. The collapse literature — the Illusion-of-Thinking exchange and the SAT-hardness evaluations [1], [9] — photographed the typhoon; we hoist numbered signals. The benchmark-reliability complaints are the motivation, and they have been shouting into the wind for years, poor souls. The one methodological addition worth naming: certificate scoring on satisfiable-only instances, which buys a two-parameter curve an honest zero floor.

6 What we do not claim

No mechanism. No physics vocabulary — nobody's water is freezing, nothing is critical, there is no order parameter hiding in section five. No entropy tea-leaves, and no claim that this live trace policy gives useful early warning. Whether trace contents can grade **individual answers** stopped being future work when we checked (Section 4.7: real signal, cannot route); the frozen persistent trigger then failed against a matched-rate timer by escalating almost everything. Whether any of this connects to a mechanism is somebody else's paper. The restraint is the brand.

7 Limitations

Two task families, one gateway, six-to-eight models: the instrument is demonstrated, not exhausted. Satisfiable-only conditioning makes high-α instances atypical of random 3-SAT, as stated. The marginal slope a compresses instance heterogeneity into one number by design. Effort in tokens is a proxy paid for at the provider's meter, and providers that hide or bill thinking differently will move it. And a $300 experiment measures what a $300 experiment measures; the generators and the runner are released precisely so that anyone with a bigger ledger can stretch every axis.

Reproducibility. Generators, raw per-call records (tokens, cost, provider endpoint, full response text), fitting code, and the exact configs that produced every figure are at github.com/posix4e/difficulty-response-curves. `analysis/run_analysis.py` regenerates every number; the abstract reads them from `numbers.json` at compile time. Git revision: `6c53b828`.

Appendix

7.1 Model and decoding settings

model string	provider pin	API	tokens	$/1M in	$/1M out
`openai/gpt-oss-20b`	parasail	chat completions	cap 32768	0.044	0.22
`openai/gpt-oss-120b`	cerebras	chat completions	cap 32768	0.385	0.825
`minimax/minimax-m2.5`	parasail	chat completions	cap 32768	0.33	1.32
`qwen/qwen3-235b-a22b-thinking-2507`	novita	chat completions	cap 32768	0.33	3.3
`deepseek/deepseek-r1-0528`	novita	chat completions	cap 32768	0.77	2.75
`openai/o4-mini`	openai	chat completions	cap 32768	1.21	4.84
`anthropic/claude-haiku-4.5`	unpinned	Anthropic Messages	thinking 8192	1.1	5.5
`z-ai/glm-5`	unpinned	chat completions	cap 32768	0.66	2.112
`openai/gpt-5.5`	openai	chat completions	cap 32768	5.5	33.0
`anthropic/claude-fable-5`	anthropic	chat completions	cap 32768	9.9	49.5
`or/claude-fable-5`	unpinned	chat completions	cap 65536	9.9	49.5
`or/gpt-5.5`	unpinned	chat completions	cap 65536	5.5	33.0
`or/glm-5`	unpinned	chat completions	cap 65536	0.6	1.92
`or/grok-4-fast`	unpinned	chat completions	cap 65536	0.2	0.5
`or/minimax-m2.5`	parasail	chat completions	cap 65536	0.3	1.2
`or/deepseek-r1-0528`	unpinned	chat completions	cap 32768	0.77	2.75

7.2 Traces: delivery by model and route

Measured from this project's own calls: of the reasoning text each call was billed for, the share that arrived (chars received over billed tokens × 3.7). Below 2 per cent means the thinking was withheld and only the final answer (plus the count) came back.

model	route	billed tok/call	delivered
`minimax-m2.5`	parasail/prepaid	29234	77%
`deepseek-r1-0528`	novita/prepaid	13799	63%
`qwen3-235b-a22b-thinking-2507`	novita/prepaid	19521	61%
`claude-haiku-4.5`	anthropic-messages	6817	6%
`claude-fable-5`	anthropic/prepaid	11169	3%
`claude-fable-5`	openrouter/Anthropic	35908	1%
`gpt-5.5`	openai/prepaid	9330	0%
`o4-mini`	openai/prepaid	9966	0%
`gpt-5.5`	openrouter/OpenAI	22833	0%
`gpt-oss-120b`	cerebras/prepaid	13838	0%
`gpt-oss-120b`	novita/prepaid	12932	0%
`gpt-oss-20b`	novita/prepaid	19481	0%
`minimax-m2.5`	novita/prepaid	25538	0%
`gpt-oss-20b`	parasail/prepaid	28395	0%

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